

SIMATS ENGINEERING

ASSESSMENT TOOL 2

COURSE NAME: Database Management Systems

COURSE CODE: CSA05

COURSE OUTCOME COVERED

CO2: *Apply the relational model, relational algebra, SQL query structures, and views to solve database querying and data manipulation problems. (BL3, BL6)*

Activity Tool : Scenario-Based Task (High)

Assessment Tool Weightage: 35%

Dr.G.Ayyappan

QUESTIONS			Total Marks: 50
Q.No	Question	Bloom's Level	Marks
1	Given the scenario of a School Management System, identify relations and write SQL to list all students with their class teacher and grade.	BL3 – Apply	5
2	A hospital wants to track patient appointments. Design the relational schema and write SQL to find doctors with more than 10 appointments this month.	BL3 – Apply	5
3	In an e-commerce scenario, write SQL queries using sub-queries to find products that have never been ordered.	BL3 – Apply	5
4	For a university database, write relational algebra expressions to find students enrolled in both 'DBMS' and 'OS' courses.	BL3 – Apply	5
5	Given a payroll scenario, create a view showing employees whose salary is below department average and write a query to update their salary by 10%.	BL6 – Create	5
6	In a banking scenario, apply JOIN operations to retrieve customer account details, transaction history, and branch information.	BL3 – Apply	5

7	A retail store needs daily sales summary. Write SQL with GROUP BY and HAVING to report total sales per category exceeding Rs. 10,000.	BL3 – Apply	5
8	For a library system scenario, construct tuple relational calculus expressions to find members who borrowed books from all genres.	BL3 – Apply	5
9	Given an inventory management scenario, define CHECK and FOREIGN KEY constraints and demonstrate their enforcement through SQL.	BL3 – Apply	5
10	Design a relational schema for an HR system and create materialized views for monthly attendance and performance summaries.	BL6 – Create	5

Code of Conduct:

I C. Sravanthi /192525463 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: C.Sravanthi

RUBRICS FOR CALCULATION:

Scenario based assessment					
Criteria	Weightage (%)	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Understanding of Scenario	20%	Demonstrates clear and complete understanding of the scenario and context	Shows good understanding with minor gaps	Partial understanding; misses key aspects	Misinterprets or fails to understand the scenario
Identification of Key Issues	20%	Accurately identifies all relevant issues and factors involved	Identifies most key issues with minor omissions	Identifies limited issues; some irrelevant points	Unable to identify key issues
Application of Concepts	25%	Applies appropriate concepts effectively to	Applies concepts with minor errors	Limited or partially correct application	Unable to apply relevant concepts

		address the scenario			
Decision Making	20%	Makes well-reasoned, logical decisions with strong rationale	Makes reasonable decisions with some justification	Makes weak decisions with limited justification	No clear decisions made or rationale provided
Clarity of Response	15%	Response is clear, structured, and logically presented	Generally clear with minor issues	Somewhat unclear or poorly structured	Disorganized and difficult to understand

ANSWERS:

1. School Management System

Relations:

STUDENT(StudentID, StudentName, ClassID, Grade)
 CLASS(ClassID, ClassName, TeacherID)
 TEACHER(TeacherID, TeacherName)

SQL Query:

```
SELECT S.StudentName, T.TeacherName, S.Grade
FROM STUDENT S
JOIN CLASS C ON S.ClassID = C.ClassID
JOIN TEACHER T ON C.TeacherID = T.TeacherID;
```

2. Hospital Appointment System

Relations:

PATIENT(PatientID, PatientName)
 DOCTOR(DoctorID, DoctorName)
 APPOINTMENT(AppointmentID, PatientID, DoctorID, AppointmentDate)

SQL Query:

```
SELECT D.DoctorID, D.DoctorName
FROM DOCTOR D
JOIN APPOINTMENT A ON D.DoctorID = A.DoctorID
WHERE MONTH(A.AppointmentDate) = MONTH(CURDATE())
AND YEAR(A.AppointmentDate) = YEAR(CURDATE())
GROUP BY D.DoctorID, D.DoctorName
HAVING COUNT(*) > 10;
```

3. E-Commerce

```
SELECT ProductID, ProductName
FROM PRODUCT
WHERE ProductID NOT IN
(
    SELECT ProductID
    FROM ORDER_DETAILS
);
```

4. University Database (Relational Algebra)

Find students enrolled in both DBMS and OS.

$$\pi_{\text{StudentID}}(\sigma_{\text{Course}='DBMS'}(\text{ENROLLMENT})) \\ \cap \\ \pi_{\text{StudentID}}(\sigma_{\text{Course}='OS'}(\text{ENROLLMENT}))$$

5. Payroll Scenario

Create View:

```
CREATE VIEW LowSalaryEmployees AS
SELECT E.EmpID, E.EmpName, E.DeptID, E.Salary
FROM EMPLOYEE E
WHERE Salary <
(
    SELECT AVG(Salary)
    FROM EMPLOYEE
    WHERE DeptID = E.DeptID
);
```

Update Salary by 10%

```
UPDATE EMPLOYEE
SET Salary = Salary * 1.10
WHERE EmpID IN
(
    SELECT EmpID
    FROM LowSalaryEmployees
);
```

6. Banking Scenario

```
SELECT C.CustomerName,
A.AccountNumber,
```

```

T.TransactionID,
T.Amount,
B.BranchName
FROM CUSTOMER C
JOIN ACCOUNT A ON C.CustomerID = A.CustomerID
JOIN TRANSACTION T ON A.AccountNumber = T.AccountNumber
JOIN BRANCH B ON A.BranchID = B.BranchID;

```

7. Retail Store

```

SELECT Category,
SUM(SalesAmount) AS TotalSales
FROM SALES
GROUP BY Category
HAVING SUM(SalesAmount) > 10000;

```

8. Library System (Tuple Relational Calculus)

```

{ M | Member(M) ∧
∀ G (Genre(G) → ∃ B ∃ L
(Book(B) ∧ Borrow(L) ∧
L.MemberID = M.MemberID ∧
L.BookID = B.BookID ∧
B.Genre = G.GenreName))
}

```

9. Inventory Management

Create Table

```

CREATE TABLE PRODUCT
(
ProductID INT PRIMARY KEY,
ProductName VARCHAR(50),
Quantity INT CHECK (Quantity >= 0),
SupplierID INT,
FOREIGN KEY (SupplierID)
REFERENCES SUPPLIER(SupplierID)
);

```

Example Enforcement

```

INSERT INTO PRODUCT
VALUES(101,'Laptop',-5,1);

```

Result:

```

ERROR:
CHECK constraint violated because Quantity cannot be negative.

```

10. HR System

Schema

EMPLOYEE(EmpID, EmpName, DeptID)

ATTENDANCE(EmpID, AttendanceDate, Status)

PERFORMANCE(EmpID, Month, Rating)

Materialized View – Attendance

```
CREATE MATERIALIZED VIEW MonthlyAttendance AS
SELECT EmpID,
COUNT(*) AS DaysPresent
FROM ATTENDANCE
GROUP BY EmpID;
```

Materialized View – Performance

```
CREATE MATERIALIZED VIEW MonthlyPerformance AS
SELECT EmpID,
AVG(Rating) AS AverageRating
FROM PERFORMANCE
GROUP BY EmpID;
```